

- 17). // C++ program to Swap Two Numbers Using Concept of
//i). Pass By value
//ii). Pass By Reference
//iii). Pass By Pointer

```
#include <iostream>
using namespace std;

// i) Pass by Value
void swapByValue(int a, int b) {
    int temp = a;
    a = b;
    b = temp;
    cout << "Inside swapByValue \n First Number: " << a << "\n Second Number:" << b << endl;
}

// ii) Pass by Reference
void swapByReference(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
    cout << "Inside swapByReference \n First Number:" << a << "\n Second Number: " << b << endl;
}

// iii) Pass by Pointer
void swapByPointer(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
    cout << "Inside swapByPointer \n First Number:" << *a << "\n Second Number: " << *b << endl;
}

int main () {
    int x,y;
    cout<<"Enter 2 Integer to Swap:"<<endl;
    cin>>x>>y;
    cout << "Original values \n First Number: " << x << "\n Second Number:" << y <<"\n"<< endl;

    // Pass by Value
    swapByValue(x, y);
    cout << "After swapByValue \n First Number: " << x << "\n Second Number:" << y << endl << endl;

    // Pass by Reference
    swapByReference(x, y);
    cout << "After swapByReference \n First Number:" << x << "\n Second Number:" << y << endl << endl;

    // Pass by Pointer
```

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```
swapByPointer(&x, &y);  
cout << "After swapByPointer \n First Number: " << x << "\n Second Number:" << y << endl;  
  
return 0;  
}
```

★ Output:



```
E:\C Programming\C++\Swap X + v  
Enter 2 Integer to Swap:  
2  
6  
Original values  
First Number: 2  
Second Number:6  
  
Inside swapByValue  
First Number: 6  
Second Number:2  
After swapByValue  
First Number: 2  
Second Number:6  
  
Inside swapByReference  
First Number:6  
Second Number: 2  
After swapByReference  
First Number:6  
Second Number:2  
  
Inside swapByPointer  
First Number:2  
Second Number: 6  
After swapByPointer  
First Number: 2  
Second Number:6  
  
-----  
Process exited after 18.96 seconds with return value 0  
Press any key to continue . . . |
```

- 18). // C++ program to Find Simple Interest Using Concept of
//i). Return By value
//ii). Return By Reference
//iii). Return By Pointer

```
#include <iostream>
using namespace std;

// i) Return by Value
float Value(float p, float r, float t) {
    float si = (p * r * t) / 100;
    return si;
}

// ii) Return by Reference
float siRef;
float& Reference(float p, float r, float t) {
    siRef = (p * r * t) / 100;
    return siRef;
}

// iii) Return by Pointer (using static variable)
float* Pointer(float p, float r, float t) {
    static float si;
    si = (p * r * t) / 100;
    return &si;
}

int main () {
    float p, r, t;
    cout << "Enter the Principal, Rate and Time (in years): ";
    cin >> p >> r >> t;
    cout << "\nPrincipal: " << p << ", Rate: " << r << ", Time: " << t << endl;

    float valueResult = Value(p, r, t);
    cout << "Simple Interest (Return by Value): " << valueResult << endl;

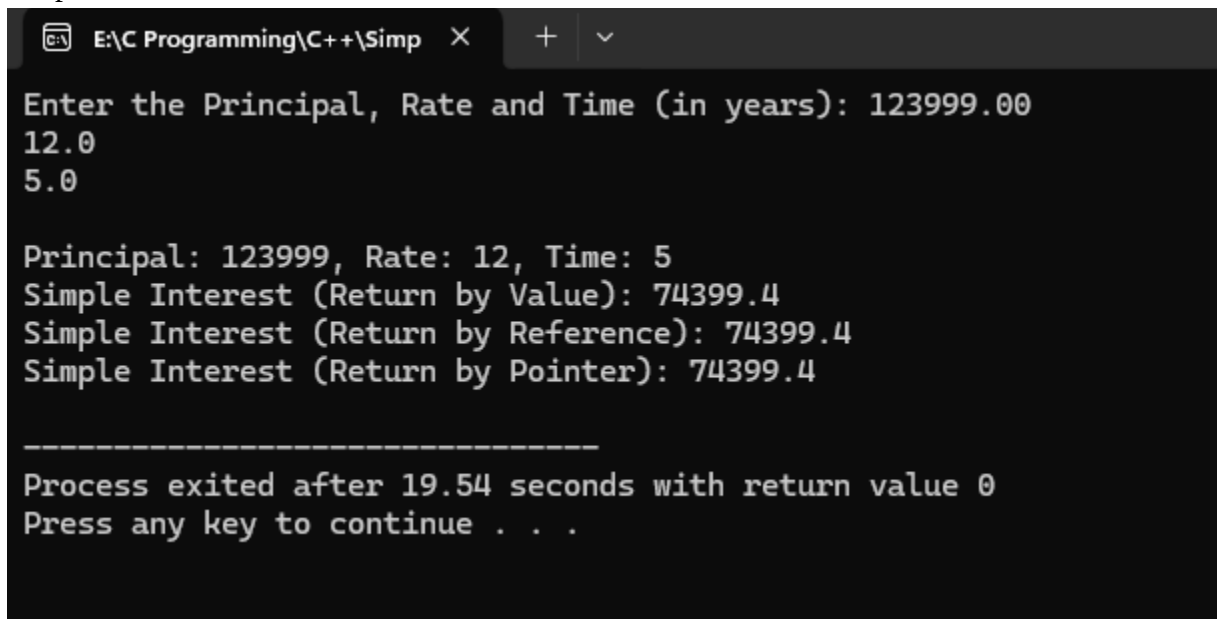
    float& referenceResult = Reference(p, r, t);
    cout << "Simple Interest (Return by Reference): " << referenceResult << endl;

    float* pointerResult = Pointer(p, r, t);
    cout << "Simple Interest (Return by Pointer): " << *pointerResult << endl;
```

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```
return 0;  
}
```

★ Output:



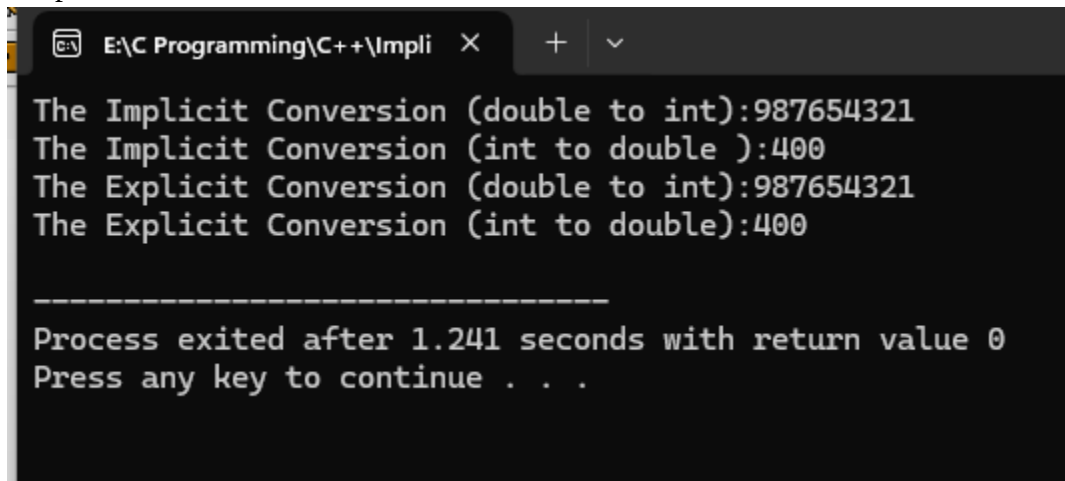
```
E:\C Programming\C++\Simp X + v  
Enter the Principal, Rate and Time (in years): 123999.00  
12.0  
5.0  
  
Principal: 123999, Rate: 12, Time: 5  
Simple Interest (Return by Value): 74399.4  
Simple Interest (Return by Reference): 74399.4  
Simple Interest (Return by Pointer): 74399.4  
  
-----  
Process exited after 19.54 seconds with return value 0  
Press any key to continue . . .
```

19). // C++ Simple Program Showing Implicit and Explicit Types conversion

```
#include<iostream>
using namespace std;

int main(){
    int a=400;
    double b= 987654321.234;
// Implicit Type Conversion
    int c=b;
    double d=a;
    cout<<"The Implicit Conversion (double to int):" <<c<<endl;
    cout<<"The Implicit Conversion (int to double ):" <<d<<endl;
//Explicit Type Conversion
    int x =(int)b;
    double y =double (a);
    cout<<"The Explicit Conversion (double to int):" <<x<<endl;
    cout<<"The Explicit Conversion (int to double):" <<y<<endl;
    return 0;
}
```

★ Output:

A screenshot of a Windows command prompt window with a dark background. The title bar shows the file path 'E:\C Programming\C++\Impli' and standard window controls. The output of the C++ program is displayed in white text. It shows four lines of output corresponding to the four cout statements in the code: 'The Implicit Conversion (double to int):987654321', 'The Implicit Conversion (int to double):400', 'The Explicit Conversion (double to int):987654321', and 'The Explicit Conversion (int to double):400'. Below these, a separator line of dashes is shown, followed by the message 'Process exited after 1.241 seconds with return value 0' and 'Press any key to continue . . .'.

```
E:\C Programming\C++\Impli X + v
The Implicit Conversion (double to int):987654321
The Implicit Conversion (int to double ):400
The Explicit Conversion (double to int):987654321
The Explicit Conversion (int to double):400

-----
Process exited after 1.241 seconds with return value 0
Press any key to continue . . .
```